

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

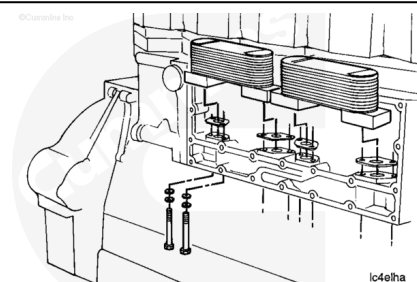
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Remove the oil cooler cover. Refer to Procedure 007-045 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-045-tr.html)
- Remove the torque converter cooler, if equipped. Refer to Procedure 008-065 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-065-tr.html)
- Remove the marine gear cooler, if equipped. Refer to Procedure 008-041 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-041.html)

Remove

Note : There is little clearance between the capscrews and the block on some engines. If necessary, use a drift and a mallet to tap the loosened capscrews from the oil cooler housing.

Remove the eight capscrews and the two oil cooler elements.

Remove and discard the gaskets.



Pressure Test

Note : Cummins, Inc. does **not** recommend any form of cleaning of oil cooler elements (internal or external). This includes cleaning with solvent, flushing with oil, steam cleaning, and ultrasonic cleaning.

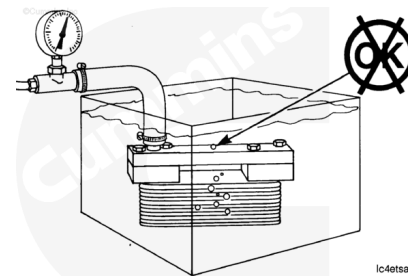
Discard the cores if debris from a malfunction is found in the oil filters.

If the engine has experienced a malfunction that allows debris to pass through the lubrication system to the full-flow oil filters and oil passages resulting in the lubricating oil cooler become contaminated, the oil cooler elements **must** be replaced.

Inspect the oil cooler elements externally for scale build-up, corrosion, embedded debris, and/or any damage to the elements. If there is any evidence of any of these conditions, the oil elements **must** be replaced.

Note : Contamination can be defined by metal, rubber, or other solid particles present in the lubricating oil filter and/or by debris damage to the bearings, or debris damage of any other components such as they do **not** meet reuse guidelines. Remove gasket material from the oil cooler cover, oil cooler element, and the engine block mating surface.

Pressure test the cores to check for leaks.



Measurements

	kpa	psi
Air Pressure	415	60

Note : Heating the water in the tank to 50°C [122°F] will improve the test results.

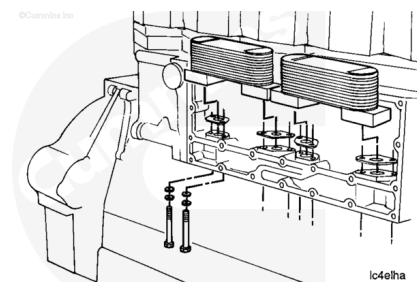
Install

Remove the plastic shipping plugs from the new oil cooler elements.

Install the gaskets, oil cooler elements, and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when filling the cooling system so that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

- Install the oil cooler cover. Refer to Procedure 007-045 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-045-tr.html)
- Install the torque converter cooler, if removed. Refer to Procedure 008-065 in Section
(/qs3/pubsys2/xml/en/procedures/20/20-008-065-tr.html) 8.
- Install the marine gear cooler, if removed. Refer to Procedure 008-041 in Section
(/qs3/pubsys2/xml/en/procedures/20/20-008-041.html) 8.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Operate the engine and check for leaks.